IT Workshop II (Cloud Computing): Cloud Concepts and Technologies, Cloud Application Design

September 7, 2026

Amazon RDS

- The input parameters to this function include the instance ID, database size, instance type, database username, database password, database port, database engine (e.g. MySQL5.1), database name, security groups, etc.
- **AWS Permissions:** We need to check if the IAM role/user running the script for RDS instance has sufficient permissions to create RDS instances and security groups.
- **Scalability:** RDS allows you to scale your database's compute resources and storage capacity with just a few clicks or API calls, depending on your application's needs.

Python Program for Launching an RDS Instance

```
import boto.rds

ACCESS_KEY="<enter>"
SECRET_KEY="<enter>"
REGION="us-east-1"
INSTANCE_TYPE="db.t1.micro"
ID = "MySQL-db-instance-3"
USERNAME = 'root'
PASSWORD = 'password'
DB_PORT = 3306
DB_SIZE = 5
DB_ENGINE = 'MySQL5.1'
DB_NAME = 'mytestdb'
SECGROUP_HANDLE="default"

#Connecting to RDS
conn =  boto.rds.connect_to_region(REGION,
  aws_access_key_id=ACCESS_KEY,
  aws_secret_access_key=SECRET_KEY)


#Creating an RDS instance
db = conn.create_dbinstance(ID, DB_SIZE, INSTANCE_TYPE,
USERNAME, PASSWORD,  port=DB_PORT, engine=DB_ENGINE,
db_name=DB_NAME, security_groups = [ SECGROUP_HANDLE, ] )
```

Value of On-Demand Services

- When owning your resources, you will pay a penalty whenever your resources do not match the instantaneous demand
  - Either pay for unused resources, or suffer the penalty of missing service delivery
- D(t) – Instantaneous Demand at time t
- R(t) – Resources at time t
- Penalty Cost $\propto \int |D(t)–R(t)|dt$

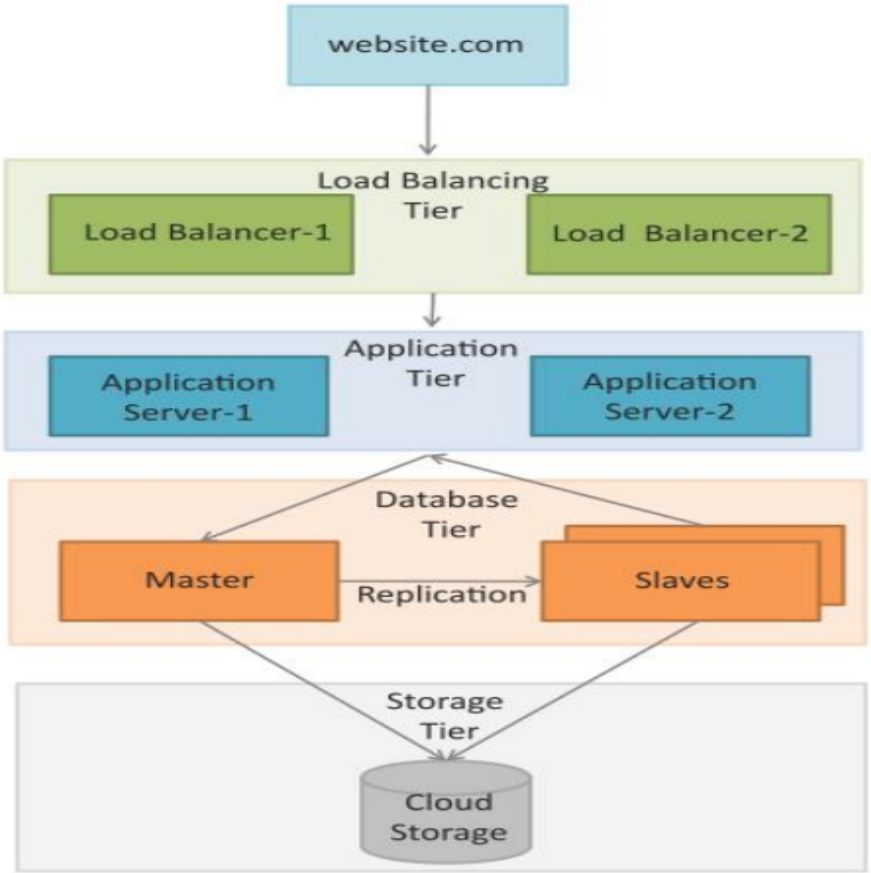
Design Considerations for Cloud Applications

- **Scalability:** Building applications that can serve millions of users without taking a hit on their performance has always been challenging.
- **Reliability & Availability:** Reliability of a system is defined as the probability that a system will perform the intended functions under stated conditions for a specified amount of time. Availability is the probability that a system will perform a specified function under given conditions at a prescribed time.

Design Considerations for Cloud Applications

- **Security:** Security is an important design consideration for cloud applications given the outsourced nature of cloud computing environments.
- **Maintenance & Upgradation:** It is important to design applications with low maintenance and upgradation costs.
- **Performance:** Applications should be designed while keeping the performance requirements in mind.

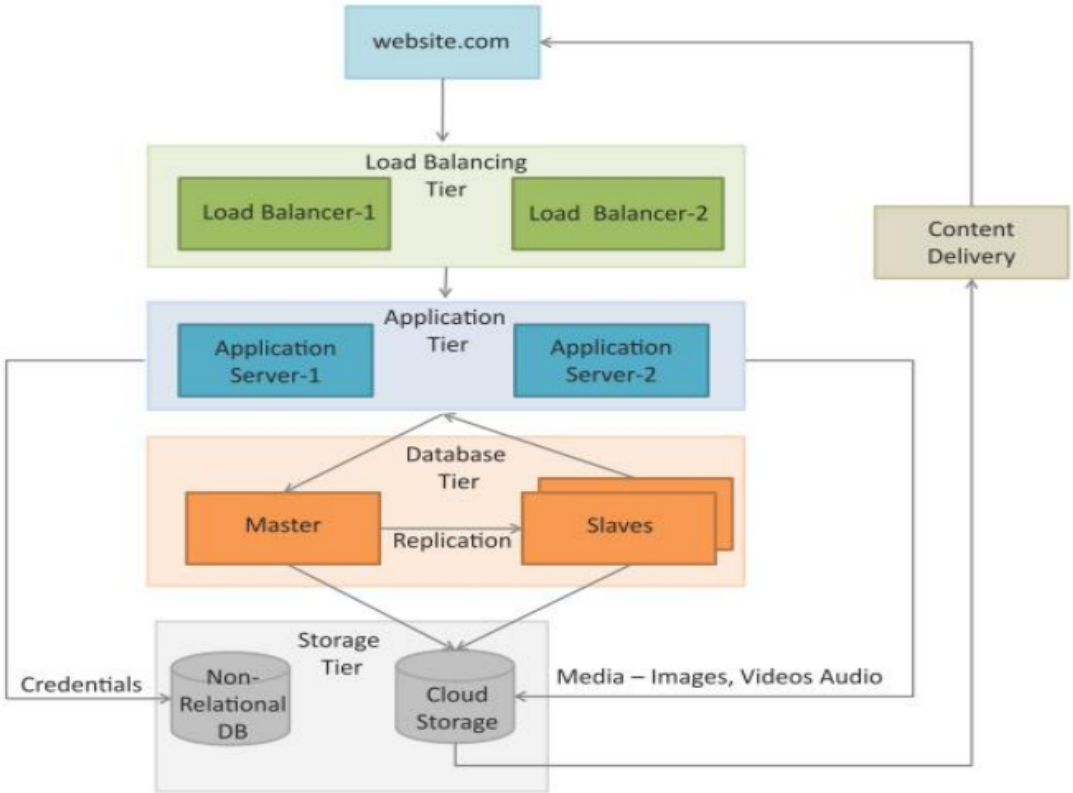
Reference Architecture - E-Commerce Application



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- **Load Balancing Tier:** Load balancing tier consists of one or more load balancers.
- **Application Tier:** For this tier, it is recommended to configure auto scaling. Auto scaling can be triggered when the recorded values for any of the specified metrics such as CPU usage, memory usage, etc. goes above defined thresholds.
- **Database Tier:** The database tier includes a master database instance and multiple slave instances. The master node serves all the write requests and the read requests are served from the slave nodes. This improves the throughput for the database tier since most applications have a higher number of read requests than write requests.

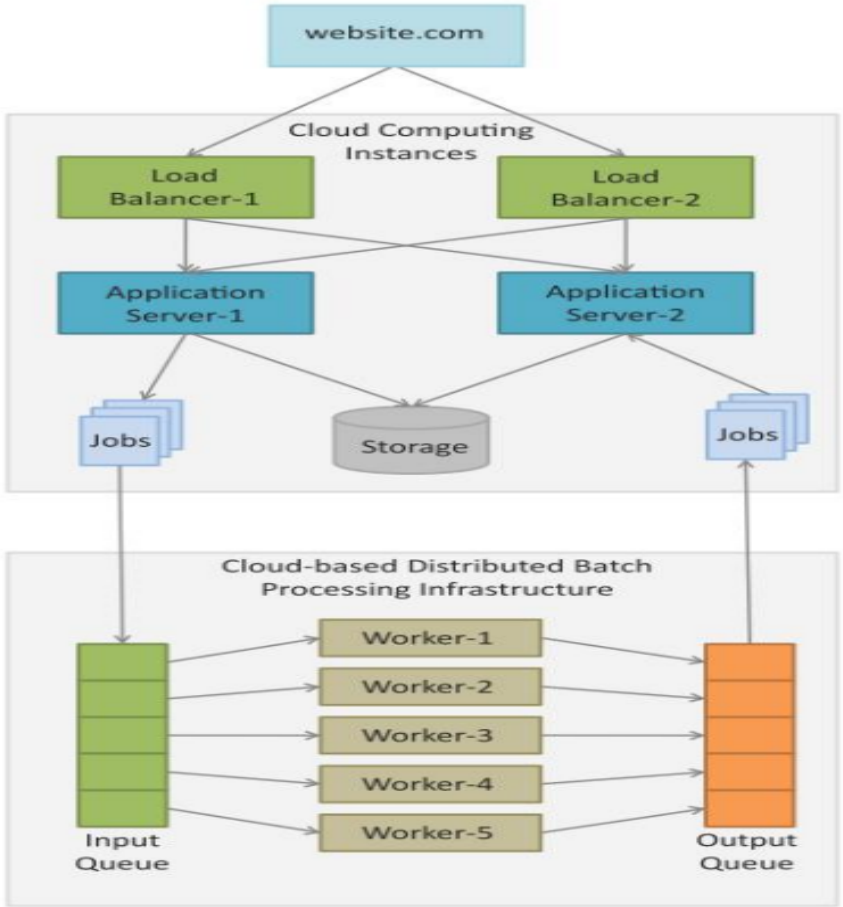
Reference Architecture - Content Delivery Apps



Reference Architecture - Content Delivery Apps

- Both relational and non-relational data stores are shown in this deployment.
- A content delivery network (CDN) which consists of a global network of edge locations is used for media delivery.
- CDN is used to speed up the delivery of static content such as images and videos.

Reference Architecture - Analytics Apps



Reference Architecture - Analytics Apps

- The architecture comprises of web, application, storage, computing/analytics and database tiers.
- The analytics tier consists of cloud-based distributed batch processing frameworks such as Hadoop which are suitable for analyzing big data.
- Data analysis jobs (such as MapReduce) jobs are submitted to the analytics tier from the application servers.
- The jobs are queued for execution and upon completion the analyzed data is presented from the application servers.

Service Oriented Architecture

- Service Oriented Architecture (SOA) is a well established architectural approach for designing and developing applications in the form services that can be shared and reused.
- SOA is a collection of discrete software modules or services that form a part of an application and collectively provide the functionality of an application.
- SOA services are developed as loosely coupled modules with no hardwired calls embedded in the services.
- The services communicate with each other by passing messages.
- Services are described using the Web Services Description Language (WSDL).
- WSDL is an XML-based web services description language that is used to create service descriptions containing information on the functions performed by a service and the inputs and outputs of the service.

Service Oriented Architecture

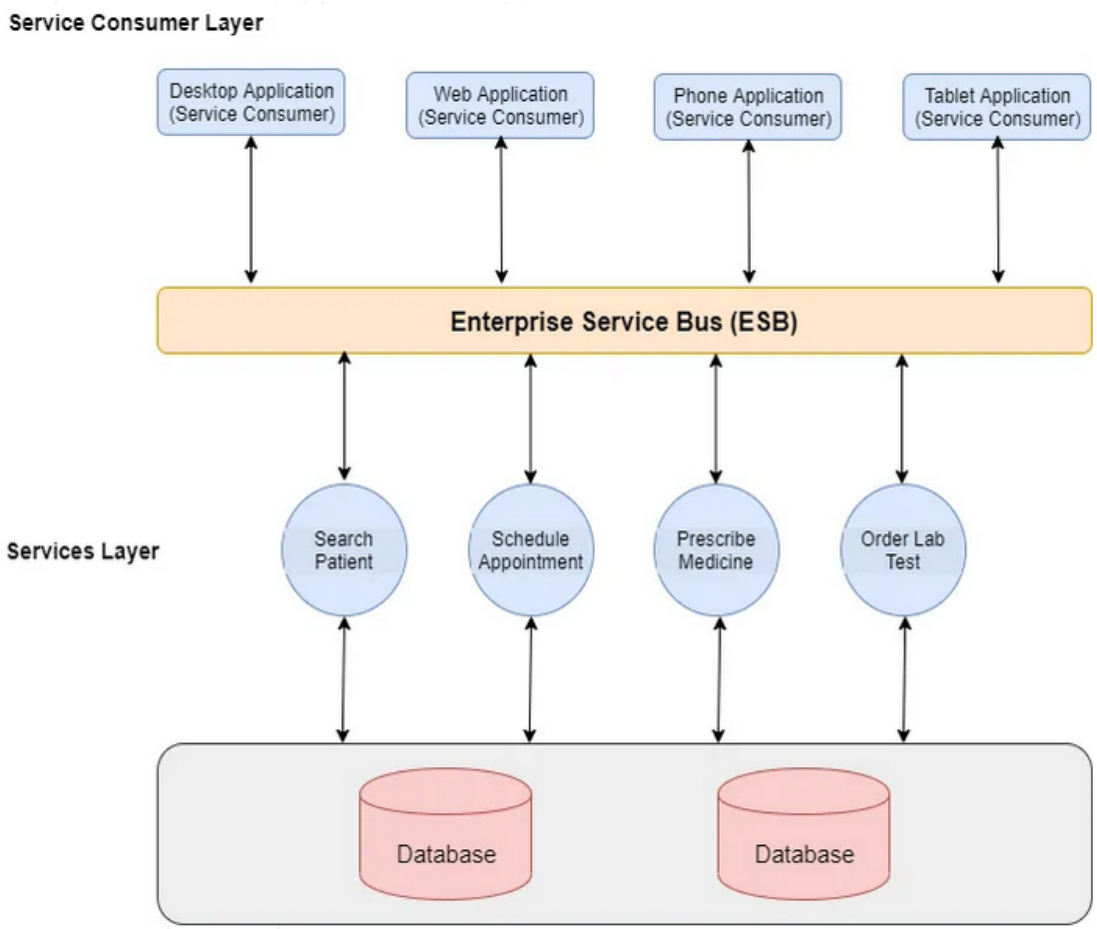


Figure: Service Oriented Architecture (Source: Internet)

The End